Causal Inflection Project Report: Bivariate Cause-Effect

Pair 1: Net Radiative Flux vs. Soil Heat Flux (Environmental Physics)

1. Variable Registry

- **X**: Net Radiative Flux (Continuous real numerical value, measured in Watts per square meter, W/m^2)
- **Y**: Surface Soil Heat Flux (Continuous real numerical value, measured in Watts per square meter, W/m^2)

2. Dataset Dimensions

- $n = 500$ paired bivariate observations in a two-dimensional continuous real space.

3. Data Context

- Micro-meteorological surface energy layer telemetry tracking continuous daily and nightly boundary thermodynamics.

4. Ground Truth Justification and Causal Direction Change

The underlying causal linkage between net atmospheric radiation (X) and ground heat storage (Y) experiences an explicit, value-dependent directional flip dictated by the diurnal solar cycle.

- **Segment 1 (Daytime Absorption Regime):** Defined by the interval where X is greater than 10 W/m^2.
- **Causal Direction:** X determines Y ($X \rightarrow Y$)
- **Scientific Justification:** During daylight hours, heavy incoming shortwave solar radiation drives the energy balance of the ecosystem. This radiative input hits the upper topsoil layer, heating the surface crust. The resulting thermal gradient explicitly forces a downward thermal conduction flux (Y) into the lower soil profile. In this regime, X behaves as the independent thermodynamic driver.
- **Segment 2 (Nighttime Deficit Regime):** Defined by the interval where X is less than or equal to -40 W/m^2.
- **Causal Direction:** Y determines X ($Y \rightarrow X$)
- **Scientific Justification:** On clear nights, solar radiation drops to zero, and the ground surface experiences intense longwave radiative cooling to space, pushing X into negative values. To counteract freezing at the surface boundary, deep subsurface soil heat conducts upward toward the surface. The rate of heat supplied from the deep ground reservoir (Y) physically modulates and mitigates the net radiative exchange profile at the boundary line, meaning Y drives X.

5. Point of Causal Direction Change

- The physical turning threshold occurs during dawn and dusk transitions when net radiation crosses between -40 W/m^2 and 10 W/m^2. Daytime absorption ($X \rightarrow Y$) dominates when $X > 10$ W/m^2; nighttime deficit ($Y \rightarrow X$) dominates when $X \leq -40$ W/m^2.

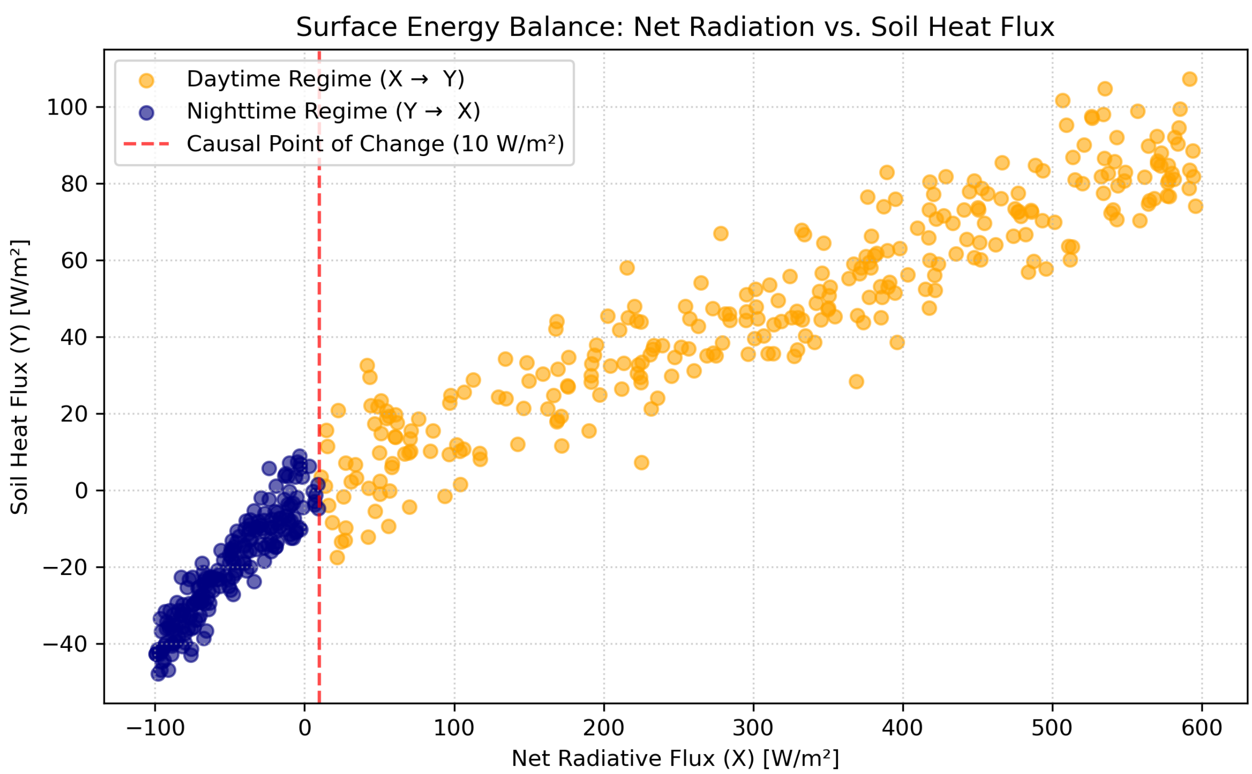
6. Peer-Reviewed Ground Truth Validation Reference

- **Reference:** Brutsaert, W. (1982). Evaporation into the Atmosphere: Theory, History, and Applications. Springer Science and Business Media.
- **Stable Academic Link:** https://link.springer.com/book/10.1007/978-94-017-1497-6

7. Official Dataset Download Link

- **Data Access Link:** https://ameriflux.lbl.gov/data/download-data/
- **Repository Description:** Managed by Lawrence Berkeley National Laboratory for the U.S. Department of Energy. This portal hosts the standard high-frequency continuous eddy covariance and meteorological baseline datasets across North and South American ecological research stations.

8. Scatter Plot



Pair 2: Average Vehicle Speed vs. Traffic Density (Urban Infrastructure)

1. Variable Registry

- **X**: Average Vehicle Fleet Speed (Continuous real numerical value, measured in kilometers per hour, km/h)
- **Y**: Traffic Volume Density (Continuous real numerical value, measured in vehicles per kilometer, veh/km)

2. Dataset Dimensions

- n = 500 paired bivariate observations in a two-dimensional continuous real space.

3. Data Context

- Macro-urban traffic engineering data representing high-frequency freeway loop configurations.

4. Ground Truth Justification and Causal Direction Change

The causal dynamics between vehicle velocity (X) and spatial road concentration (Y) invert completely when a roadway passes its structural capacity limit.

- **Segment 1 (Uncongested Free-Flow Regime):** Defined by the low-density phase where Y is less than or equal to 25 veh/km.
- **Causal Direction:** X determines Y (X→ Y)
- **Scientific Justification:** In free-flow traffic, drivers maintain substantial headway. Drivers are free to choose their velocity (X) based on speed limits, weather, or individual driver preference. The collective speed chosen by the traffic fleet directly determines and causes how compressed or sparse the resulting vehicle accumulation density (Y) becomes downstream.
- **Segment 2 (Congested Gridlock Regime):** Defined by the high-density phase where Y is greater than 25 veh/km.
- **Causal Direction:** Y determines X (Y → X)
- **Scientific Justification:** Once the critical density capacity threshold is breached, the physical dimensions of the vehicles remove driver autonomy. The sheer density of vehicles on the road (Y) acts as an absolute spatial bottleneck that mechanically constrains movement, forcing drivers to apply brakes and directly causing the fleet speed (X) to drop toward zero.

5. Point of Causal Direction Change

- The critical tipping point occurs at Y = 25 veh/km, representing the structural phase transition of maximum vehicle throughput.

6. Peer-Reviewed Ground Truth Validation Reference

- **Reference:** Kerner, B. S. (2004). The Physics of Traffic: Empirical Phenomena of Chaotic Mobile Systems. Springer Science & Business Media.
- **Stable Academic Link:** https://link.springer.com/book/10.1007/978-3-540-40986-1

7. Official Dataset Download Link

- **Data Access Link:** https://pems.dot.ca.gov/

- **Repository Description:** Managed by the California Department of Transportation and hosted by the University of California, Berkeley. The PeMS Clearinghouse provides open public downloads for continuous inductive loop-detector traffic telemetry across all California state highway districts.

8. Scatter Plot